

- This exam contains 10 pages (including the front and back of this cover page) and 15 problems. Check to see if any pages are missing.
- This is the second exam and will count for 25% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems or multiple choice problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 140 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	10	Total
Points:	22	20	16	20	18	20	12	12	140
Score:									

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

☐ 2 pts (a) $\sum_{k=0}^4 \left(-\frac{3}{2}\right)^k = \frac{55}{16}.$

☐ True☐ False

☐ 2 pts (b) The 21^{st} term of the geometric sequence with initial term π and common ratio $\frac{1}{2}$ is $\frac{\pi}{2^{20}}.$

☐ True☐ False

☐ 2 pts (c) The sum of the first four thousand terms of the geometric sequence with initial term 3 and common ratio -1 is $-3.$

☐ True☐ False

☐ 2 pts (d) A sequence beginning with 1, 3, 9, 15 could be an arithmetic sequence.

☐ True☐ False

☐ 2 pts (e) A recursive definition of the geometric sequence with initial term a and common ratio r is $a_0 = a, a_{n+1} = ra_n.$

☐ True☐ False

☐ 2 pts (f) $2n^2 + \frac{3}{n} = O(n^2).$

☐ True☐ False

☐ 2 pts (g) If $f_1(n)$ is $O(g(n))$ and $f_2(n)$ is $O(g(n))$, then $f_1(n) + f_2(n)$ is $O(2g(n))$ but not $O(g(n)).$

☐ True☐ False

☐ 2 pts (h) For all integers r, s, t , if $r < s$, then $rt < st.$

☐ True☐ False

☐ 2 pts (i) For integers r, s, t , if $r|t$ and $s|t$, then $(rs)|t.$

☐ True☐ False

☐ 2 pts (j) The set S of positive integers that end with the digit 17 can be defined recursively by:
(i) $17 \in S$, (ii) If $x \in S$, then $x + 10 \in S.$

☐ True☐ False

☐ 2 pts (k) If n is a positive integer, then $\gcd(n, 2n) = n.$

☐ True☐ False

10 pts

2. The 10^{th} term of an arithmetic sequence is -4 and the 16^{th} term is 47 . What is the 11^{th} term?

Ⓐ $-\frac{161}{2}$

Ⓑ $\frac{51}{6}$

Ⓒ $\frac{9}{2}$

Ⓓ $\frac{35}{4}$

10 pts

3. What is the 222^{nd} term in the arithmetic sequence with initial term -14 and a common difference of $\frac{3}{2}$?

Ⓐ 317.5

Ⓑ 319

Ⓒ 316

Ⓓ -345.5

4 pts

4. Mark the choice that makes the following sentence true: For all integers a, b, c , it is true that $a + (b + c) = (a + b) + c$. That is, addition is an _____ operation on the integers.

☒ **A** associative☐ B commutative☐ C distributive☐ D inductive

8 pts

5. Let a and b be integers, and suppose there exist integers s, t such that $as + bt = 4$. Mark all of the items below that are valid possibilities for the value of $\gcd(a, b)$.

☒ A 1☐ B 16☐ C 5☐ D 3

4 pts

6. Mark the choice that makes the following sentence true: There are _____ positive integers that divide 72.

☐ A 10☒ B 12☐ C 6☐ D 36

10 pts

7. Mark the choice corresponding to the first seven terms of the sequence defined recursively by $a_0 = 1$, $a_1 = 1$, and, for $n \geq 2$, $a_n = 1 + a_{n-1}a_{n-2}$.

Ⓐ 1, 1, 2, 3, 5, 22, 155

Ⓑ 1, 1, 1, 3, 7, 13, 21

Ⓒ 3, 7, 13, 21, 31, 43, 57

Ⓓ 1, 1, 2, 5, 26, 677, 458330

10 pts

8. A set, S , of strings over the alphabet $\Sigma = \{a, b, c\}$ is defined recursively by:

(i) $\lambda \in S$,

(ii) if $x \in S$ then $axbc \in S$.

Mark the choice corresponding to the strings in S of length nine or less.

Ⓐ $\lambda, a, bc, abc, aabc, abcbc, aabcbc, aaabcbc, aaabcbcbc$

Ⓑ $bc, abc, bcbc, abcbc, bcbcbc, abcbcbc, bcbcbcbc, abcbcbcbc$

Ⓒ $\lambda, abc, aabcbc, aaabcbcbc$

Ⓓ $\lambda, axbc, aaxbcbc, aaaxbcbcbc$

12 pts

9. Compute $\gcd(1452, 525)$ using the Euclidean algorithm.

6 pts

10. Convert FED_{16} to decimal and mark the choice corresponding to your answer.

Ⓐ 65232

Ⓑ 4077

Ⓒ 509.6

Ⓓ 3598

10 pts

11. Find all solutions $x \in \mathbb{Z}$ to the congruence $3x \equiv 5 \pmod{8}$.

10 pts

12. If you all you have are nickles and quarters, in how many ways can you pay an \$11 bill? For example, one way would be 5 nickles and 43 quarters.

- 12 pts 13. The Fibonacci sequence is defined recursively by

$$f_0 = 0, \quad f_1 = 1, \quad \text{and, for } n \geq 2, \quad f_n = f_{n-1} + f_{n-2}.$$

Use induction to prove that for all $n \geq 0$,

$$\sum_{k=0}^n f_k = f_{n+2} - 1.$$

12 pts

14. Prove by induction: For $n \geq 1$,

$$\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$$

5 (bonus)

15. Convert 144_7 to base 2.Ⓐ 101010_2 Ⓑ 11010_2 Ⓒ 1101_2 Ⓓ 1010001_2